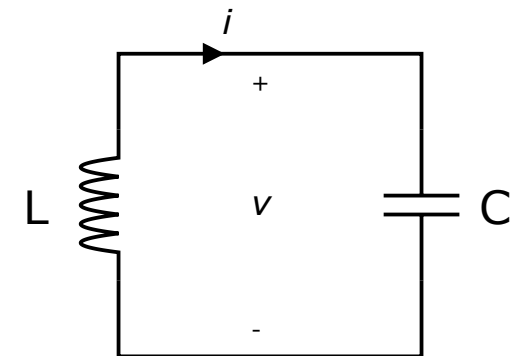
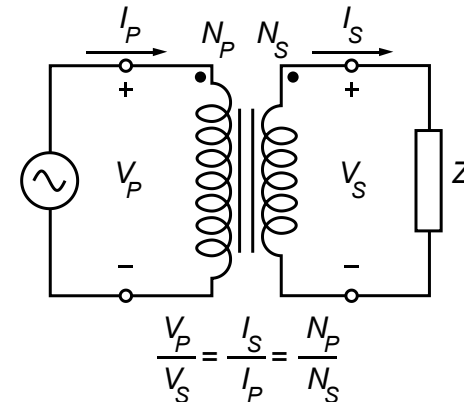
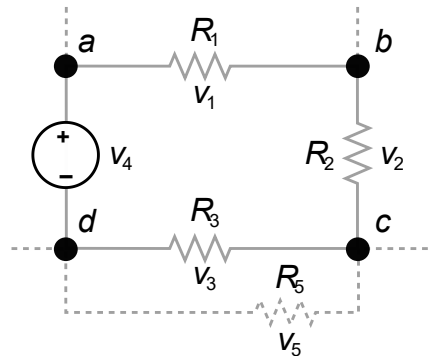
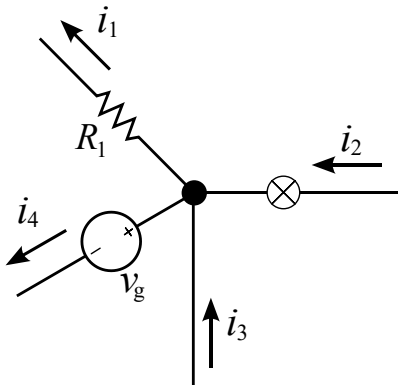


# Introduction to Electrical Engineering



# Overview

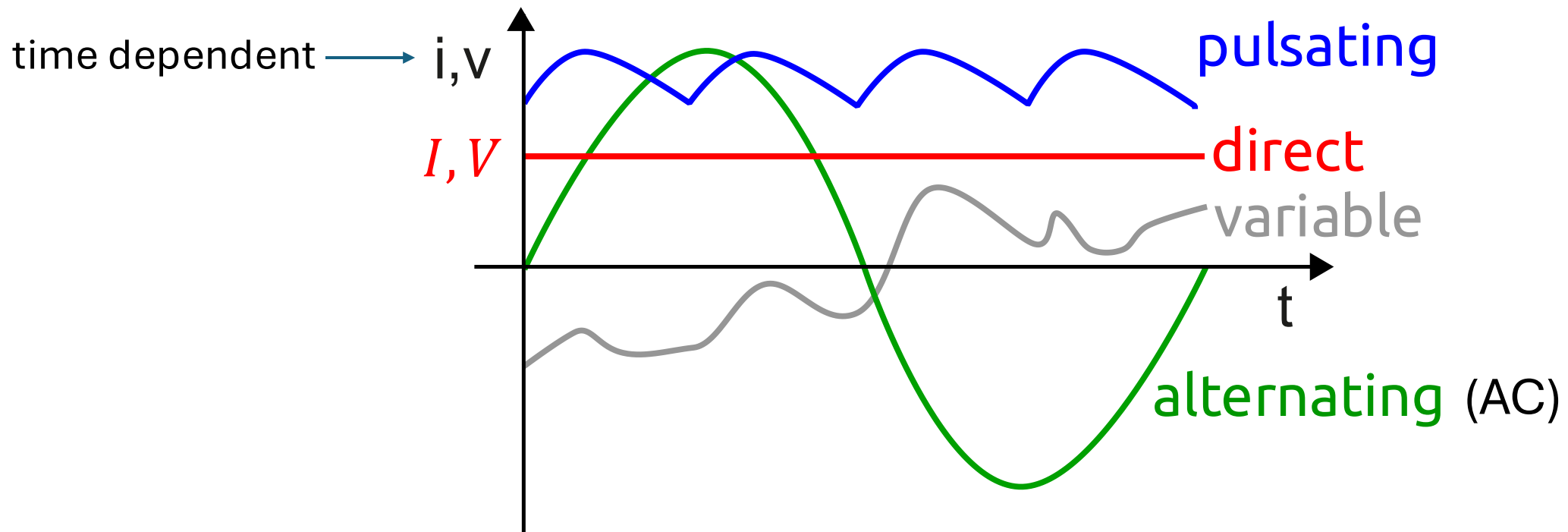
1. Physical Principles and Basic Concepts
- 2. Direct Current Circuits**
3. Electric and Magnetic Fields
4. Alternating Current Technology
5. Wrap-Up and Outlook

# Direct Current Circuits

- Reference Directions
- Kirchhoff's Current Law
- Kirchhoff's Voltage Law
- Series Circuit
- Parallel Circuit
- Practical Voltage & Current Sources
- Network Analysis
- Efficiency & Power/Impedance Matching
- Voltage & Current Measurement

# Direct Current (DC)

one-directional (typically constant) flow of electric charge



# Circuit Calculations

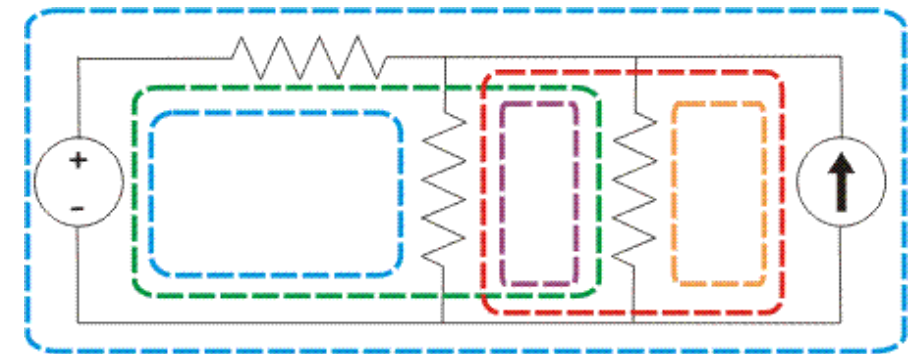
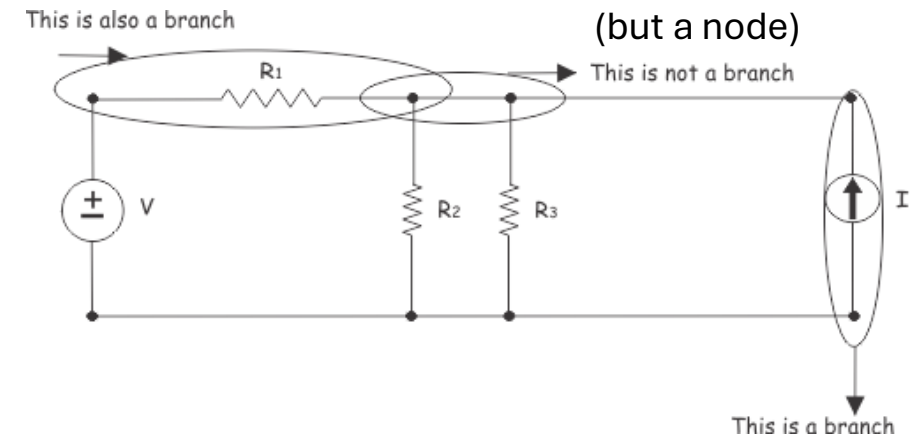
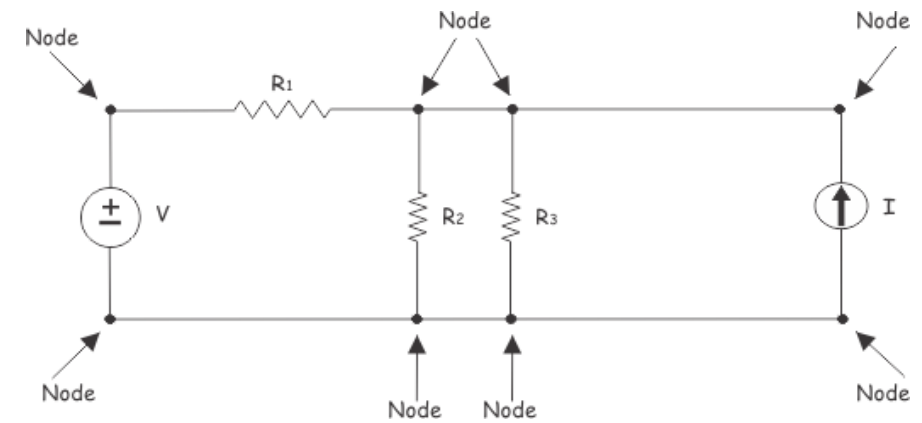
goal: calculate currents and voltages

network: interconnection of electrical components/elements (often large-scale)

circuit: closed-loop network (rather small)

fundamental concepts:

- nodes (junctions)
- branches
- loops



# Short and Open Circuits

short circuit: circuit element with resistance approaching zero

open circuit: circuit element with resistance approaching infinity

# Reference Directions

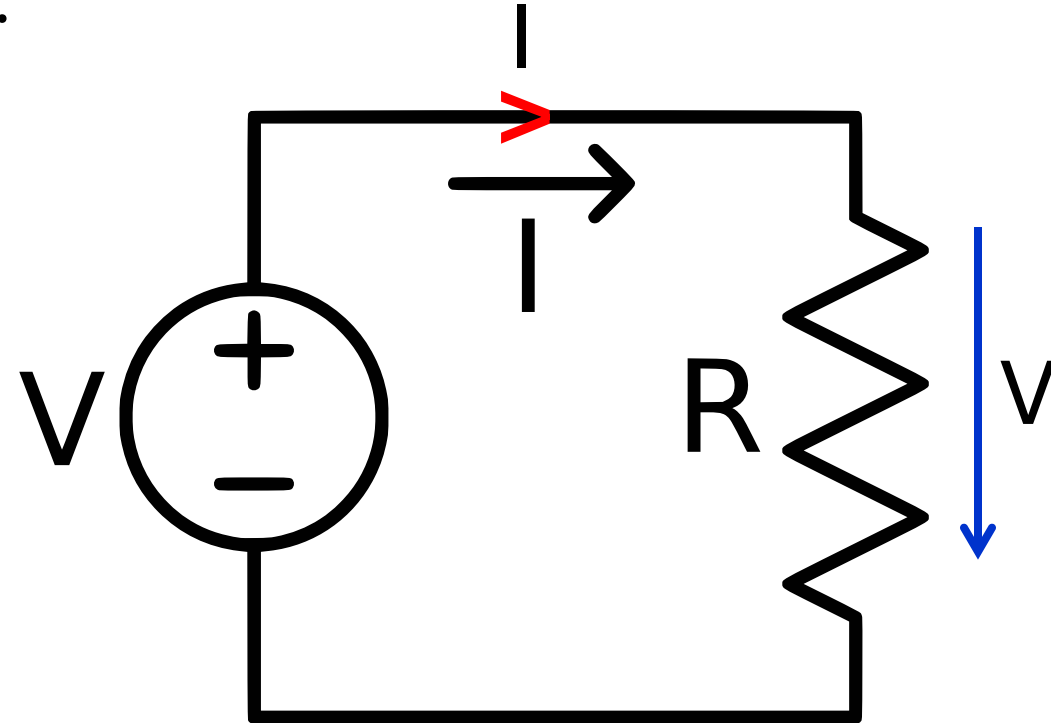
before calculation of voltages and currents:

## **arbitrary choices**

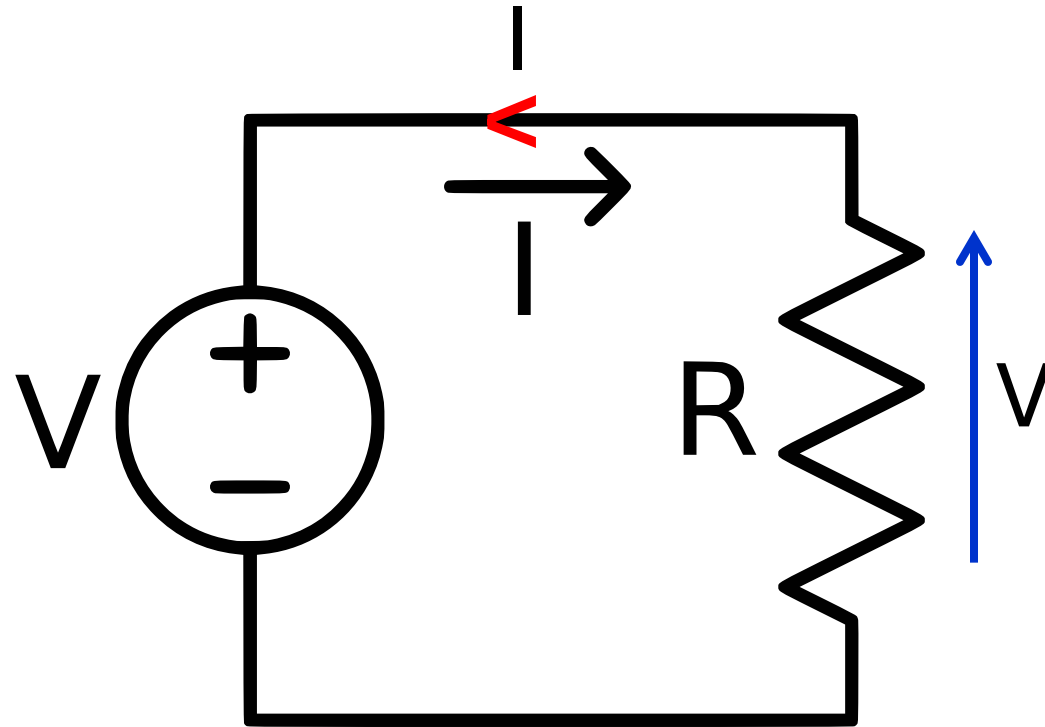
- define positive current directions
- define voltage directions + to -

after calculation:

negative value → actual direction opposite of chosen reference direction



also correct:



Will just result in negative values for  $I$  and  $V$ !

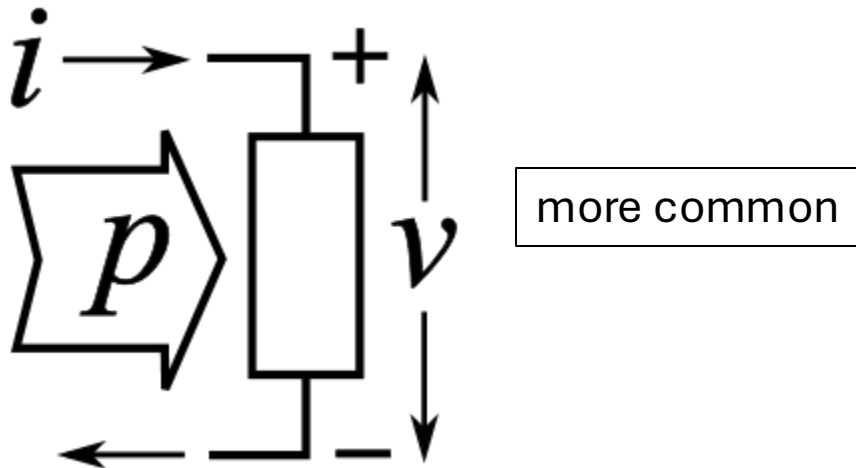


# Sign Conventions

## passive sign convention:

reference direction of current points into positive reference terminal of voltage

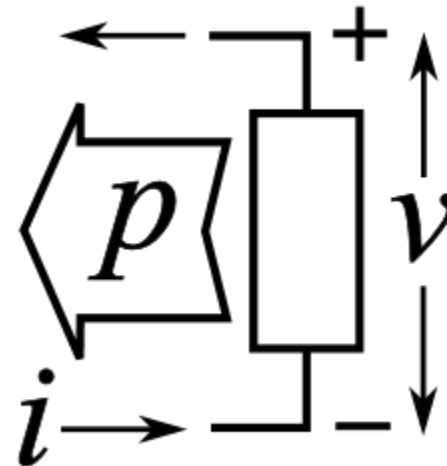
- $P > 0$  for power flowing into component (passive component)
- $P < 0$  for active component



## active sign convention:

reference direction of current points into negative reference terminal of voltage

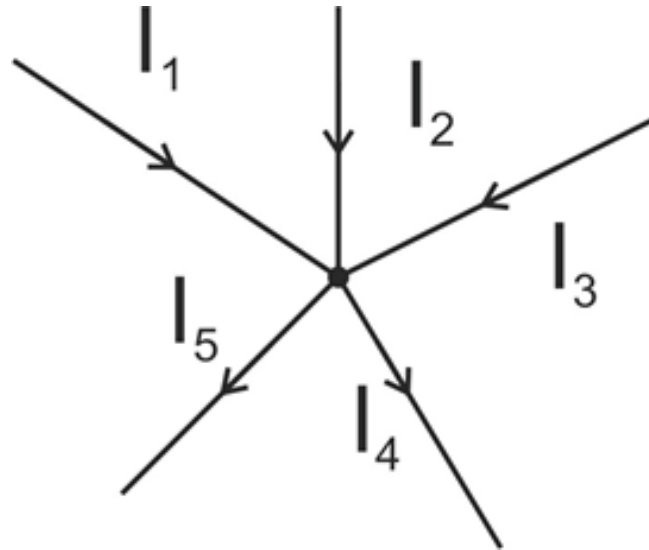
- $P > 0$  for power flowing out of component (active component)
- $P < 0$  for passive component



# Kirchhoff's Current Law (KCL)

sum of all currents entering a junction (node) equals sum of all currents leaving that junction (conservation of charge)

$$\sum_{k=1}^n I_k = 0$$

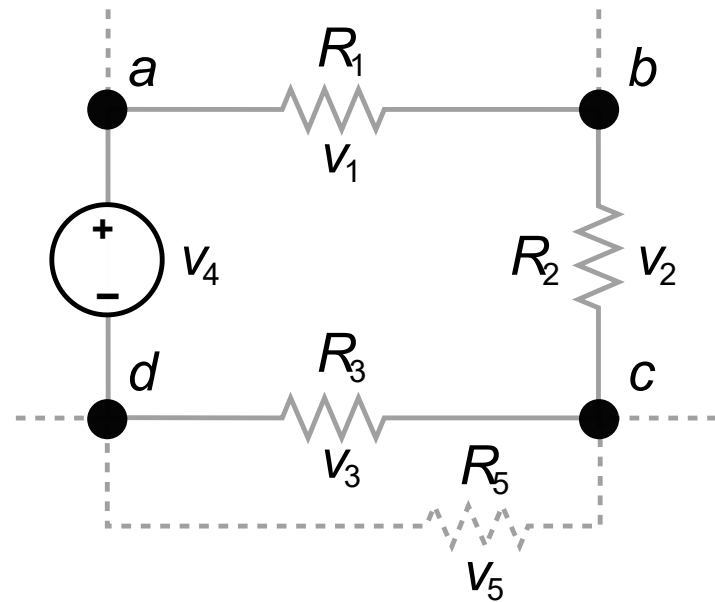


$$I_1 + I_2 + I_3 = I_4 + I_5$$

# Kirchhoff's Voltage Law (KVL)

sum of all voltage drops around any closed loop equals to zero  
(conservation of energy)

$$\sum_{k=1}^n V_k = 0$$

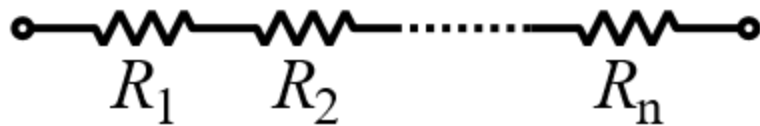
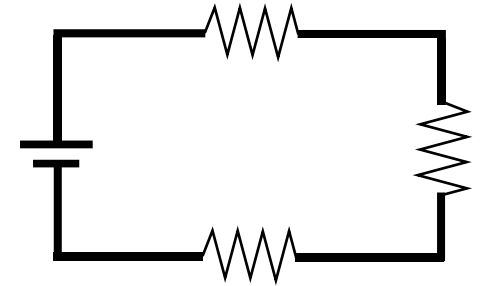


$$V_4 = V_1 + V_2 + V_3$$

# Series Circuit

- sum of potential drops around a closed loop equals supply voltage (KVL)
- same current through all components (only one path)

$$\rightarrow V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = IR$$



$$R = \sum_{k=1}^n R_k$$

voltage divider rule:

$$V_k = V \frac{R_k}{R}$$

for two resistors:

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

equivalent resistance:  
exhibits same  $i - v$  characteristics as  
original network at considered terminals

# Parallel Circuit

- sum of branch currents equals current from source (KCL)
- same voltage across each branch (KVL)

$$\rightarrow I = I_1 + I_2 + \dots + I_n = \frac{V}{R_1} + \frac{V}{R_2} + \dots + \frac{V}{R_n} = \frac{V}{R}$$

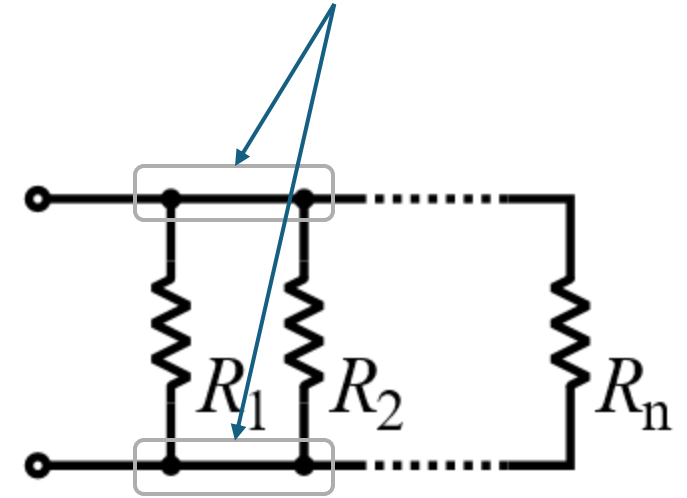
$$R = \left( \sum_{k=1}^n \frac{1}{R_k} \right)^{-1}$$

$$G = \sum_{k=1}^n G_k$$

current divider rule:  $I_k = I \frac{G_k}{G}$

for two resistors:  $\frac{I_1}{I_2} = \frac{R_2}{R_1}; \quad R = \frac{R_1 R_2}{R_1 + R_2}; \quad I_1 = I \frac{R_2}{R_1 + R_2}$

connected to the same two nodes

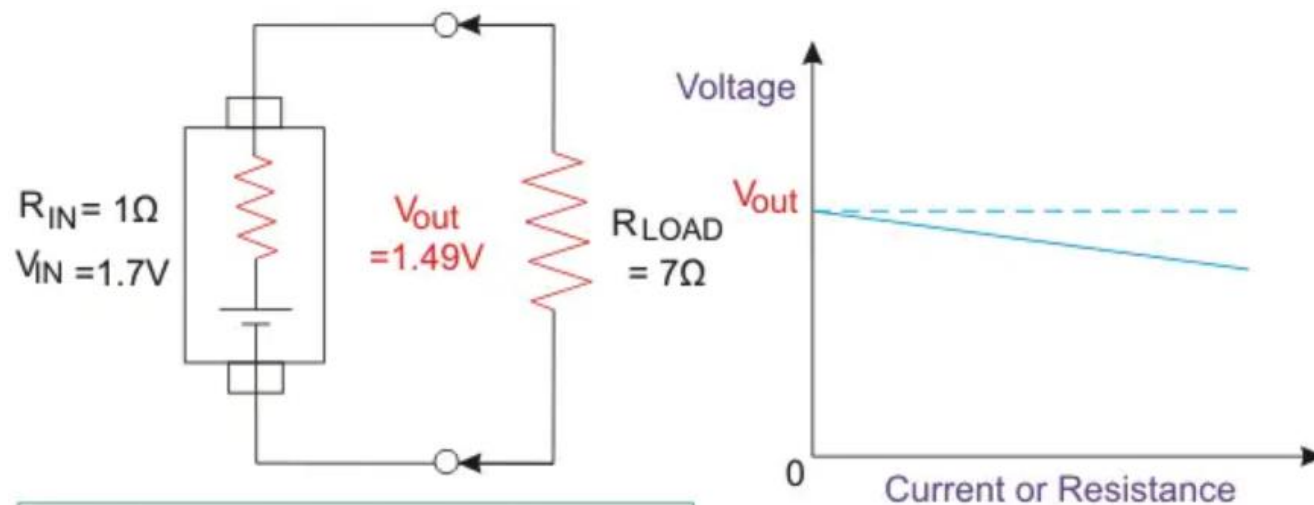


# Practical Voltage Source

equivalent circuit  
of real component

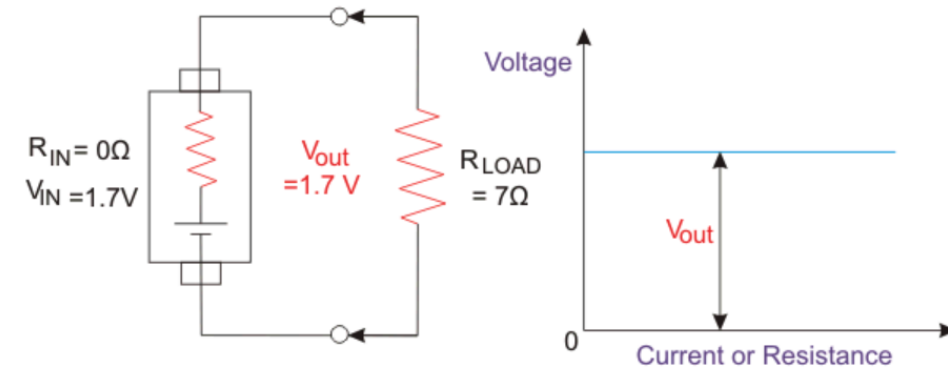
ideal voltage source in series with internal resistance

→ voltage dependent on load



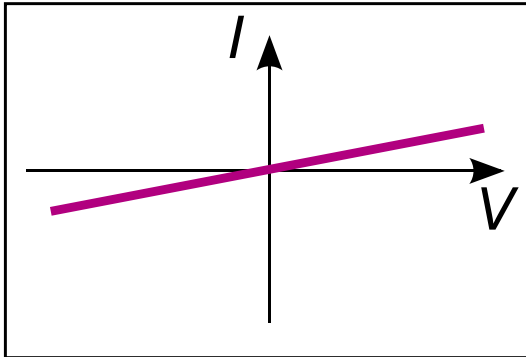
$$I = V / (R_{LOAD} + R_{IN}) = 1.7 / 8 = 0.2125A$$
$$R_{IN} \text{ Source Voltage} = I \times R_{IN} = 0.2125V$$
$$\text{So, } V_{out} = 1.7 - 0.2125 = 1.49V$$

ideal voltage source:

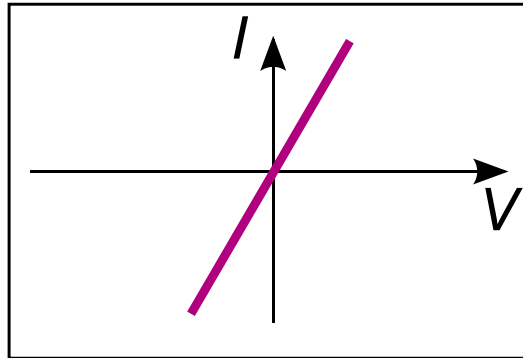


# I-V Curves

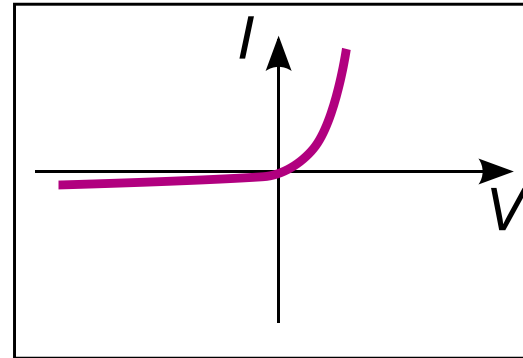
Large resistance



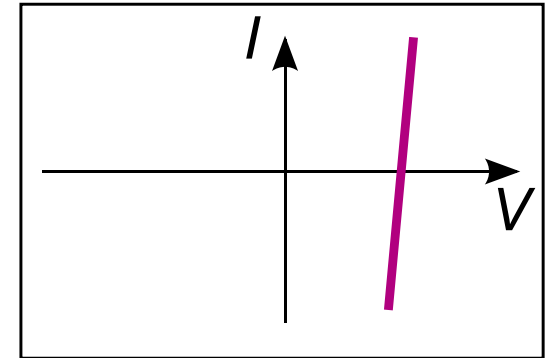
Small resistance



Diode



Battery



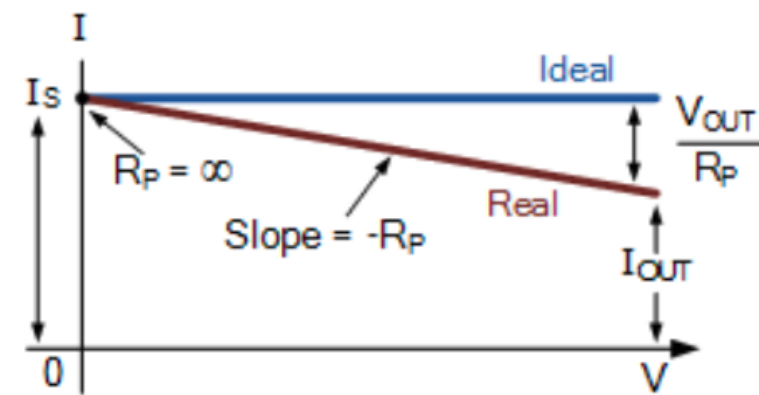
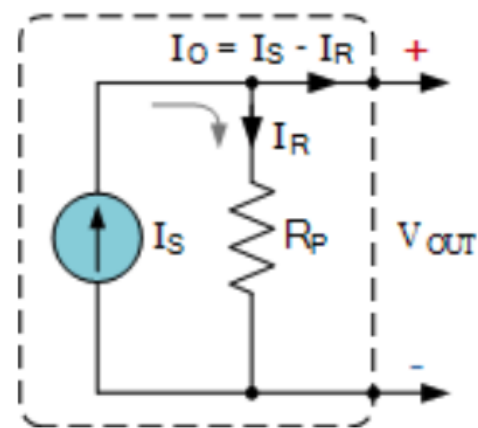
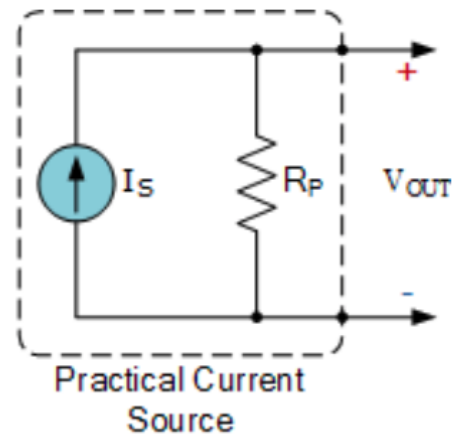
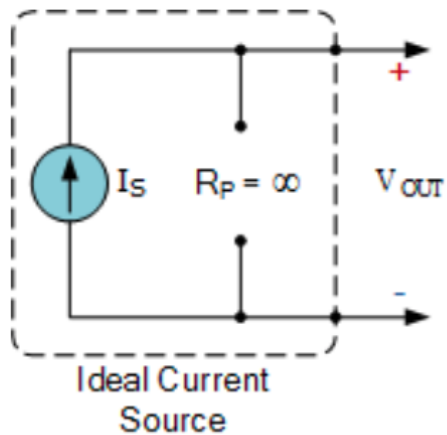
nonlinear

(considerations here only  
apply to linearized region  
around operating point)

# Practical Current Source

internal resistance in parallel with ideal current source

→ current dependent on load





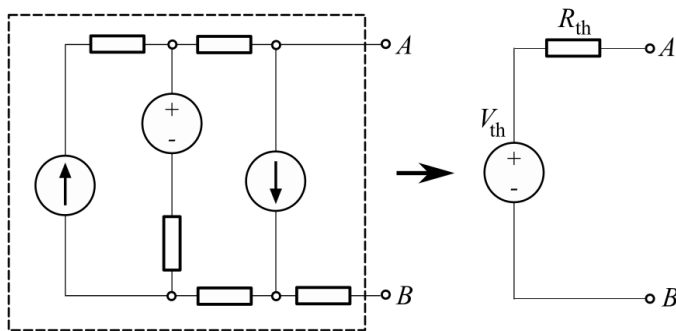
# Several Independent Sources

simplification by means of superposition principle in linear circuits:

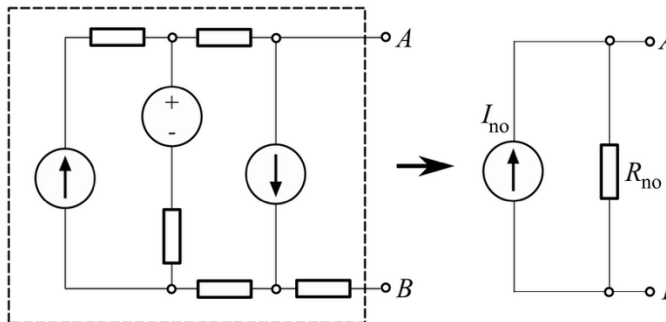
- turn on one source at a time (others turned off: replace voltage sources with short circuits and current sources with open circuits)
- find its contribution to the terminal voltage/current
- add all contributions

additionally, single “turn off all” reduction to get  $R_{th}$

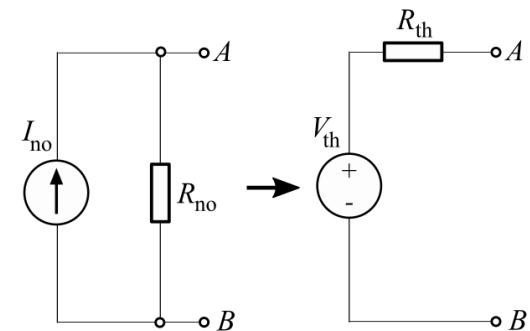
Thévenin's theorem:



Norton's theorem:



Norton-Thévenin conversion:



$$R_{th} = R_{no}$$
$$I_{no} = V_{th}/R_{th}$$

any linear circuit seen  
from two terminals

# Circuit Analysis

approach for mixed circuits:

- reduce series groups (same current path): add resistances
- reduce parallel groups (same two nodes): add conductances
- redraw after each combine, repeat

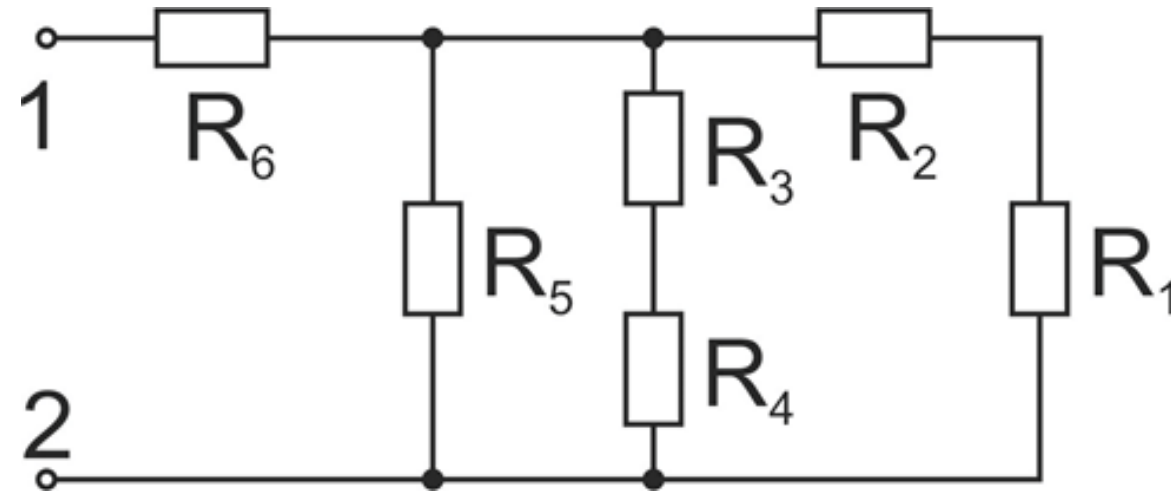
example: total resistance between 1 and 2

$$R_{1,2} = R_1 + R_2, \quad R_{3,4} = R_3 + R_4$$

$$G_{1,2,3,4} = G_{1,2} + G_{3,4}$$

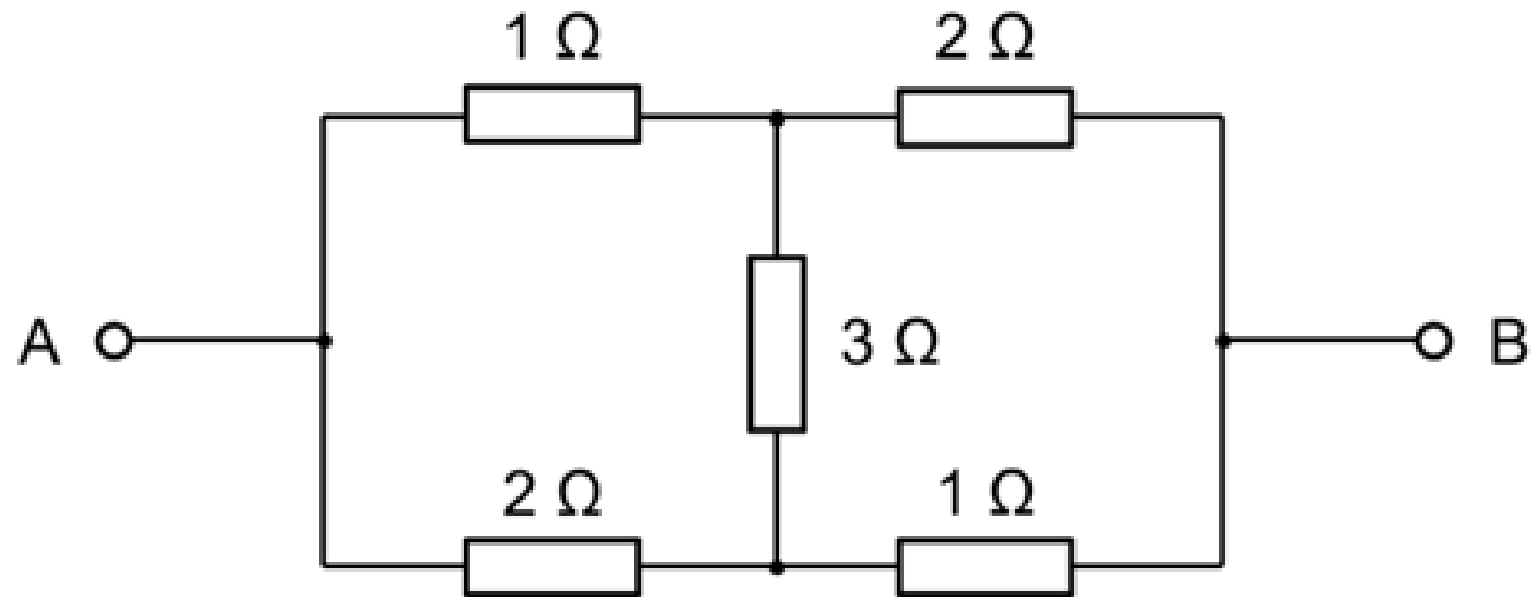
$$G_{1,2,3,4,5} = G_{1,2,3,4} + G_5$$

$$R = R_{1,2,3,4,5} + R_6$$



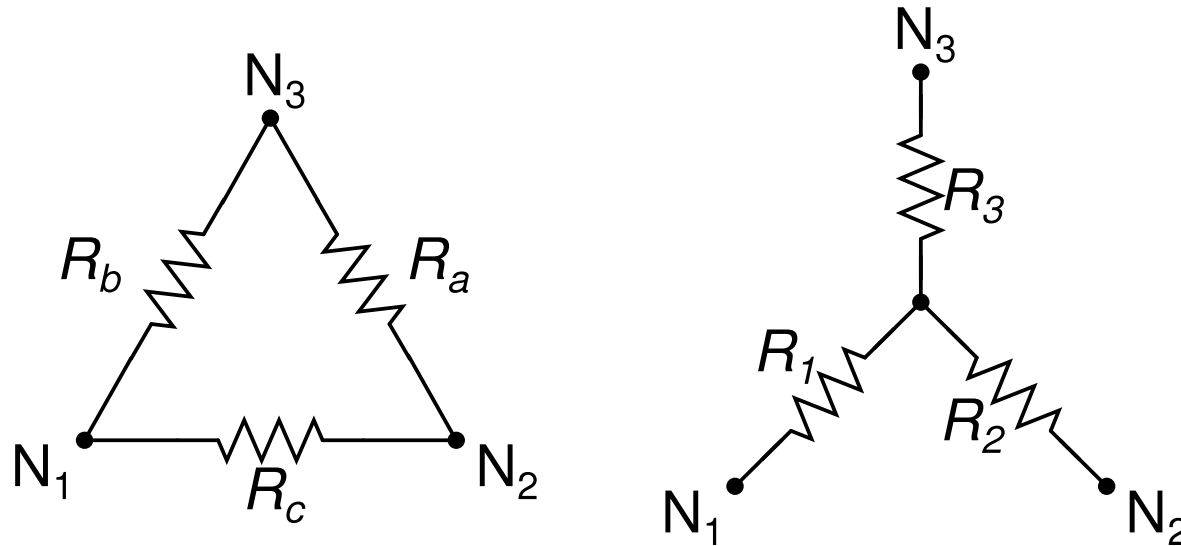
# Bridges

How to solve this one?



→ convert the bridge with Y- $\Delta$  transform  
(simplification by using three-terminal equivalent networks)

# Y-Δ Transform



idea of Y-Δ equivalence:  
define same effective resistance  
between corresponding terminals

e.g., for  $N_1$  and  $N_2$ :

$$R_1 + R_2 = \left( \frac{1}{R_c} + \frac{1}{R_a + R_b} \right)^{-1}$$

transformation from  $\Delta$  to Y:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

transformation from Y to  $\Delta$ :

$$R_a = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

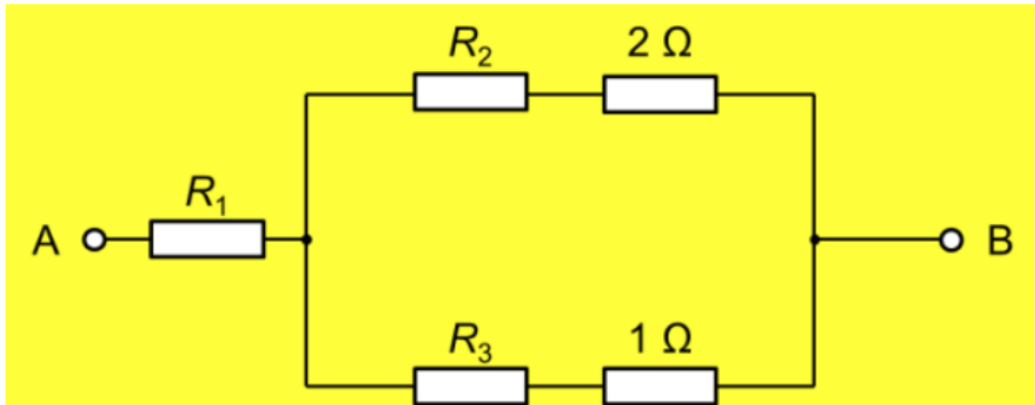
$$R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2}$$

$$R_c = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

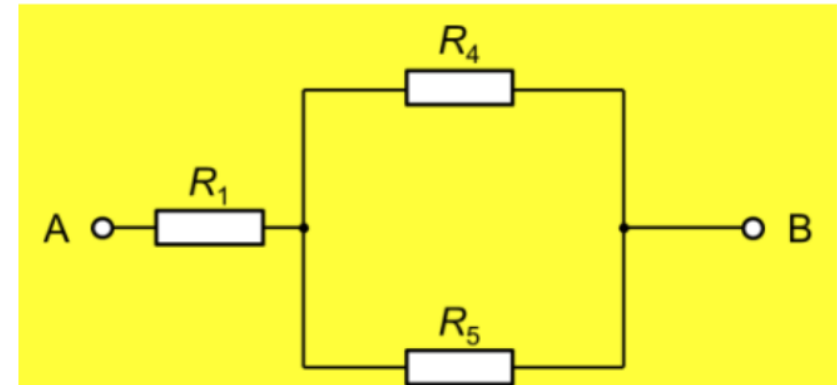
## 1. $\Delta$ to Y

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} = \frac{1 \Omega \cdot 2 \Omega}{1 \Omega + 3 \Omega + 2 \Omega} = 1/3 \Omega$$

$$R_2 = \frac{1 \Omega \cdot 3 \Omega}{1 \Omega + 3 \Omega + 2 \Omega} = 0,5 \Omega; \quad R_3 = \frac{2 \Omega \cdot 3 \Omega}{1 \Omega + 3 \Omega + 2 \Omega} = 1 \Omega$$



## 2. combine



$$R_4 = R_2 + 2 \Omega = 2,5 \Omega; \quad R_5 = R_3 + 1 \Omega = 2 \Omega$$

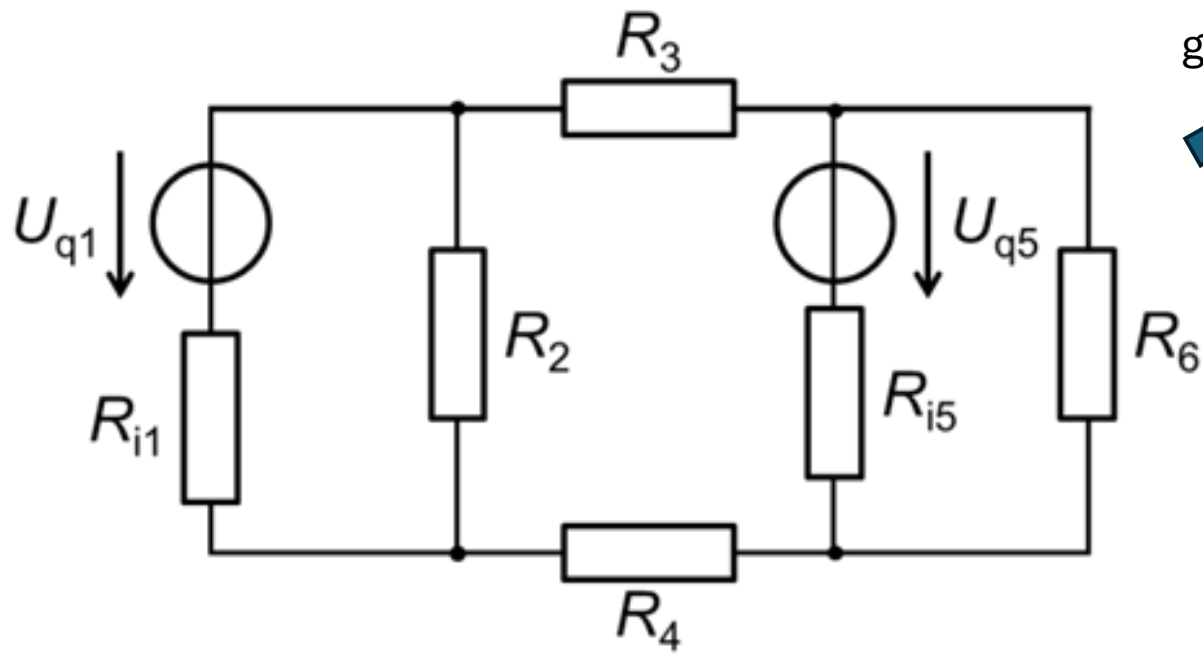
$$R_{AB} = R_1 + \frac{R_4 R_5}{R_4 + R_5} = 1/3 \Omega + \frac{2,5 \Omega \cdot 2 \Omega}{2,5 \Omega + 2 \Omega} = 1,44 \Omega$$

# Systematic Network Analysis

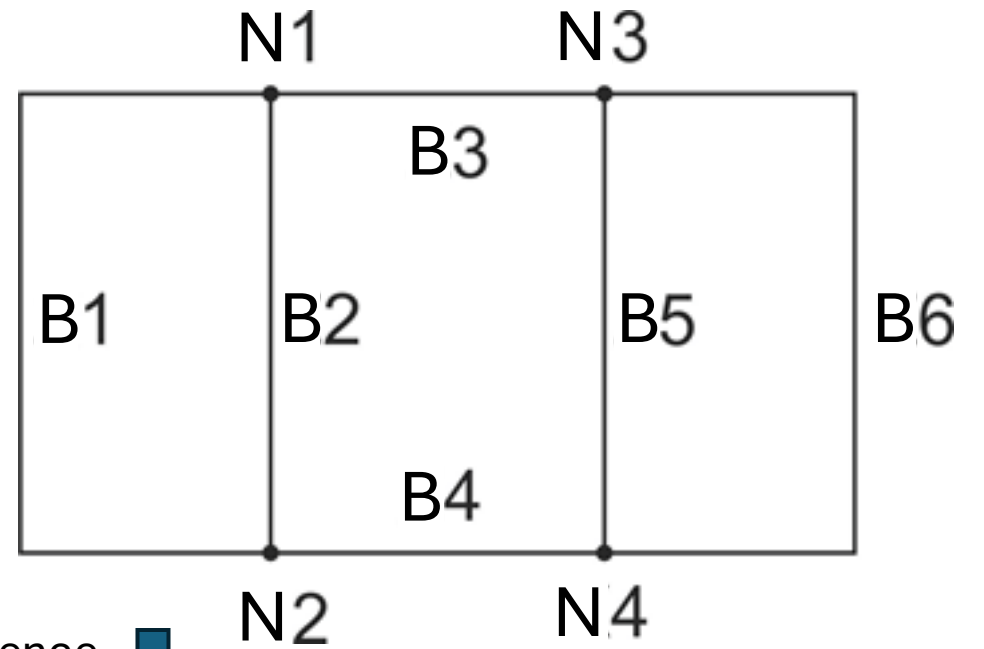
for DC resistive networks

1. represent the network
2. formulate equations: choose analysis method (different heuristics) and apply KCL / KVL / Ohm's law accordingly
3. solve the resulting system of linear equations

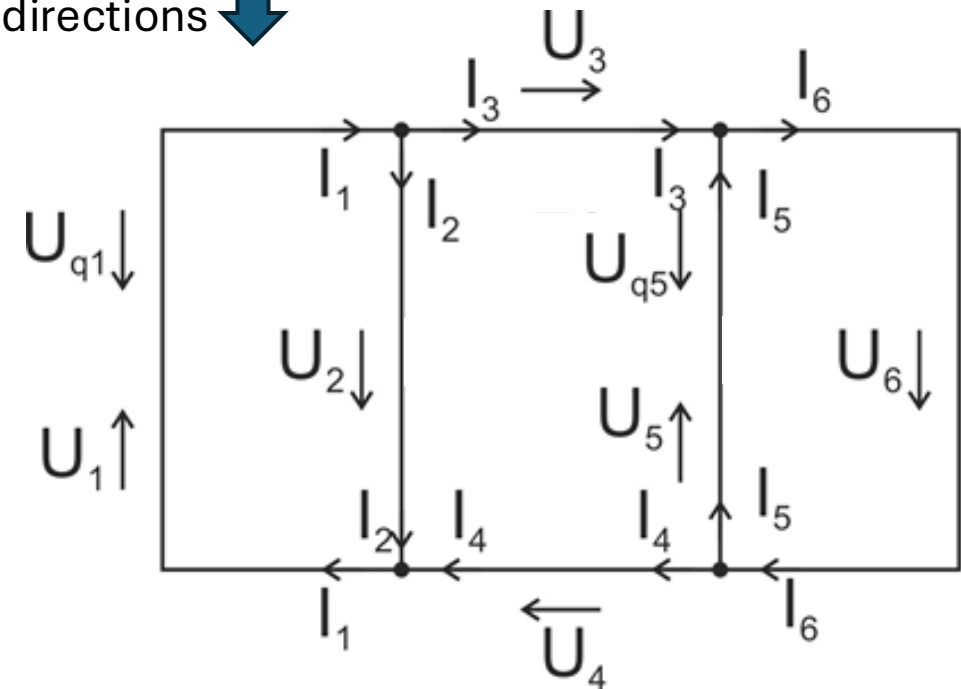
# Representation



graph



reference  
directions



# Formulation of Equations

for network with  $n$  nodes:  $n - 1$  linearly independent node equations  
here:

- N1:  $I_1 - I_2 - I_3 = 0$
- N2:  $-I_1 + I_2 + I_4 = 0$
- N3:  $I_3 + I_5 - I_6 = 0$

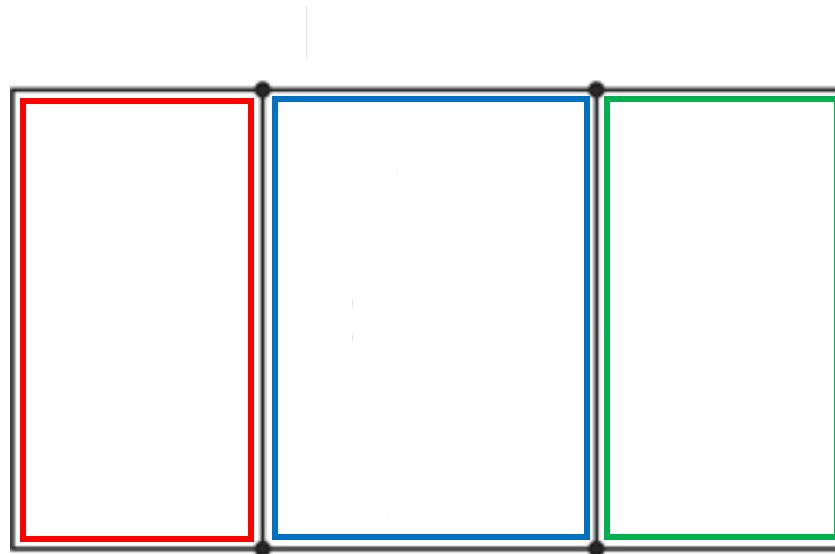
for  $b$  branches:  $m = b - (n - 1)$  linearly independent loop equations  
here:  $b = 6, n = 4 \Rightarrow m = 3 \rightarrow$  need to pick 3 out of the 6 loops



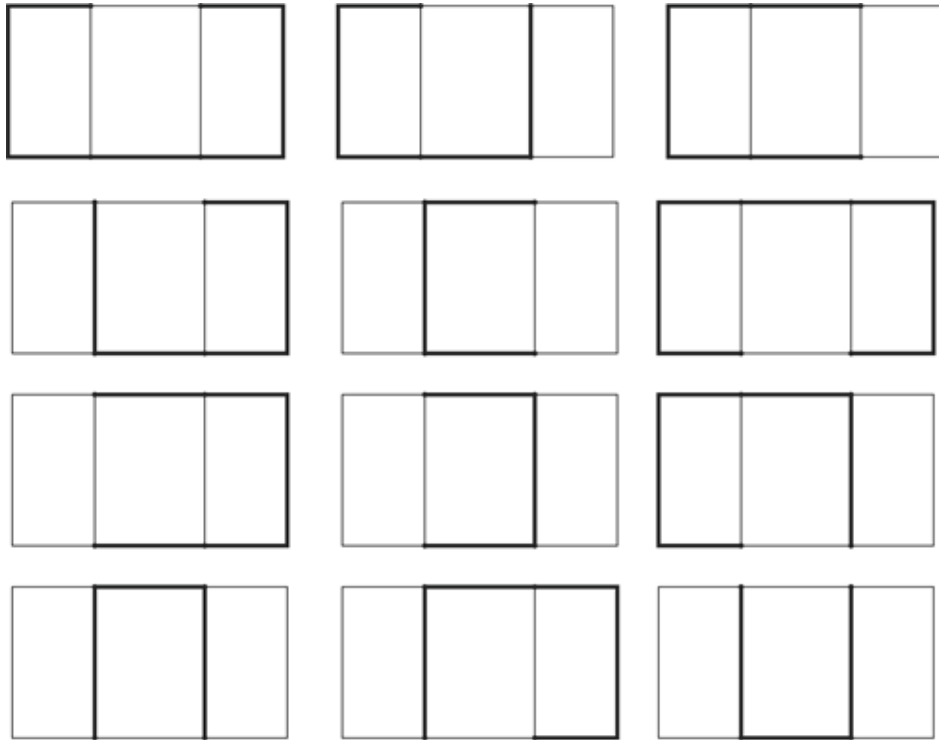
# Approach 1: Use Meshes

meshes: smallest possible loops, not enclosing any other loop

here: 3 meshes give 3 linearly independent loop equations



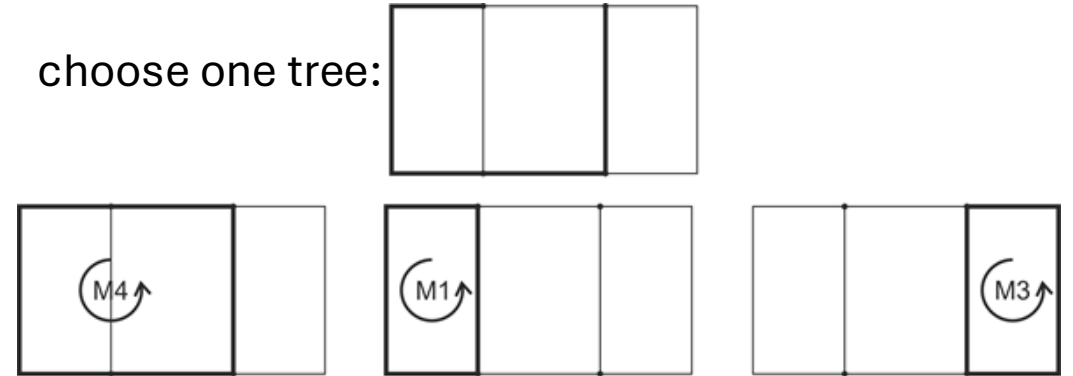
# Approach 2: Method of Complete Tree



tree: set of branches connecting all nodes  
without forming any loop

co-tree: all branches – tree branches

choose one tree:



get fundamental loops (not necessarily meshes) by  
adding the different co-tree branches back to the tree

$$\text{M1:} \quad U_1 + U_2 = U_{q1}$$

$$\text{M3:} \quad U_5 + U_6 = U_{q5}$$

$$\text{M4:} \quad U_1 + U_3 + U_4 - U_5 = U_{q1} - U_{q5}$$

# System of Linear Equations

$$I_1 - I_2 - I_3 = 0$$

$$-I_1 + I_2 + I_4 = 0$$

$$I_3 + I_5 - I_6 = 0$$

$$U_1 + U_2 = U_{q1} \Rightarrow R_{i1}I_1 + R_2I_2 = U_{q1}$$

$$U_5 + U_6 = U_{q5} \Rightarrow R_{i5}I_5 + R_6I_6 = U_{q5}$$

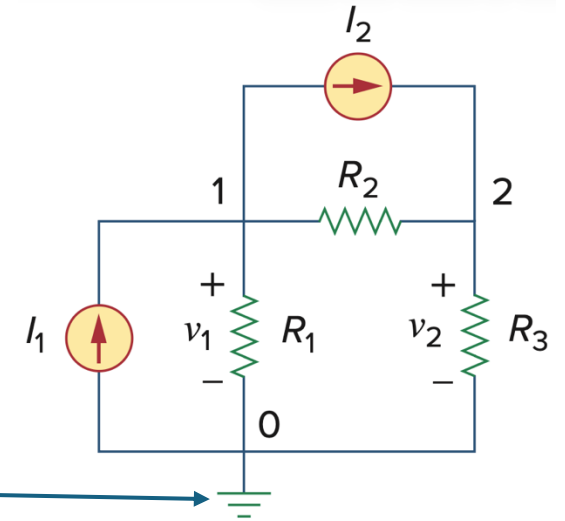
$$U_1 + U_3 + U_4 - U_5 = U_{q1} - U_{q5} \Rightarrow R_{i1}I_1 + R_3I_3 + R_4I_4 - R_{i5}I_5 = U_{q1} - U_{q5}$$

$$\begin{pmatrix} 1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ R_{i1} & R_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & R_{i5} & R_6 \\ R_{i1} & 0 & R_3 & R_4 & -R_{i5} & 0 \end{pmatrix} \cdot \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ U_{q1} \\ U_{q5} \\ U_{q1} - U_{q5} \end{pmatrix}$$

# Alternative: Nodal Analysis

sometimes simpler approach:

- select one node as reference node with an arbitrary potential (typically ground, i.e.,  $V = 0$ )



- apply KCL to  $n - 1$  non-reference nodes

$$\text{here: node1: } I_1 = I_2 + i_1 + i_2 \quad \text{node2: } I_2 + i_2 = i_3$$

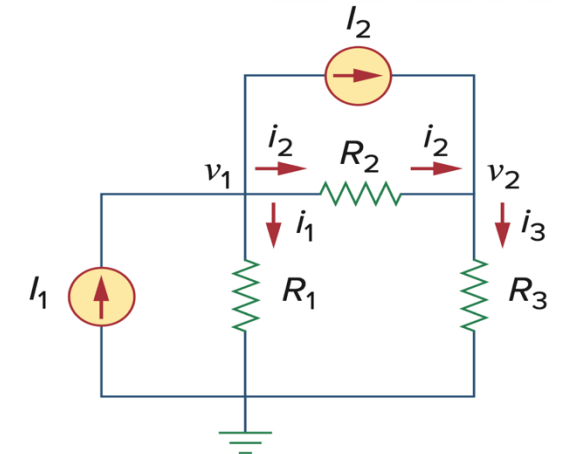
- apply Ohm's law to express currents in terms of node voltages:

$$i = \frac{v_{\text{higher}} - v_{\text{lower}}}{R}$$

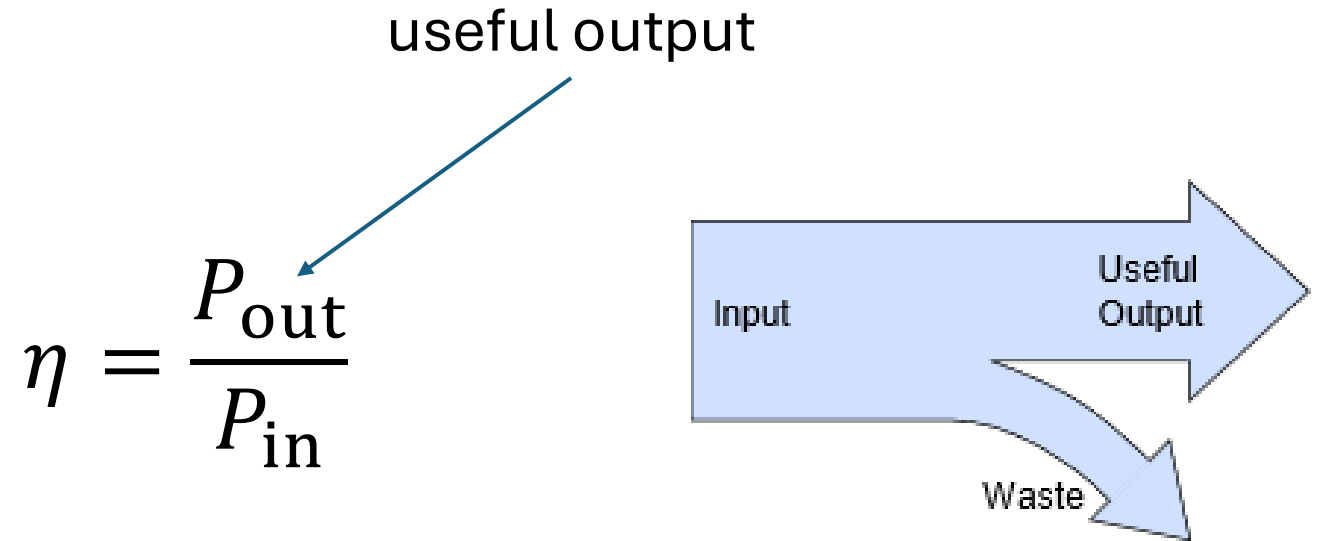
$$\text{here: } i_1 = \frac{v_1 - 0}{R_1} \quad i_2 = \frac{v_1 - v_2}{R_2} \quad i_3 = \frac{v_2 - 0}{R_3}$$

- substitute and solve

redrawn with currents:



# Efficiency



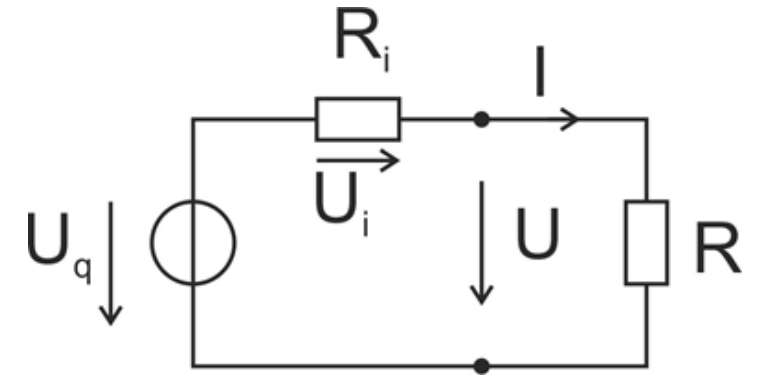
example:

A device draws 200 W and has a useful output of 120 W.

→  $\eta = 60\%$ , i.e., wasting 40% of the input power

# Power/Impedance Matching

practical voltage source with resistive load:



$$P_{\text{out}} = U \cdot I; \quad P_{\text{in}} = U_q \cdot I; \quad P_{\text{waste}} = I^2 \cdot R_i$$

$$\eta = \frac{UI}{U_q I} = \frac{U_q \frac{R_L}{R_i + R_L} \frac{U_q}{R_i + R_L}}{U_q \frac{U_q}{R_i + R_L}} = \frac{R_L}{R_i + R_L}$$

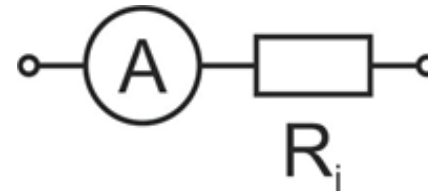
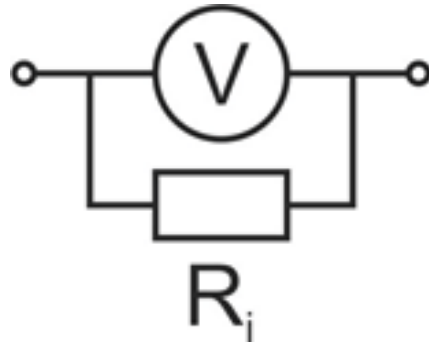
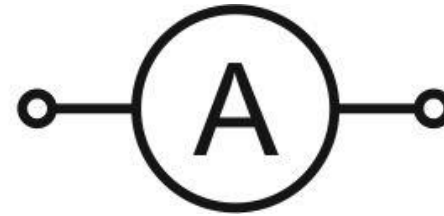
$\eta = 100\%$  at  $R_i = 0$

$\eta = 50\%$  at  $R_i = R_L$

maximum power transfer:

$$\frac{dP_{\text{out}}}{dR_L} = 0 = U_q^2 \frac{(R_i + R_L)^2 - 2R_L(R_i + R_L)}{(R_i + R_L)^4} \Leftrightarrow R_i = R_L$$

# Voltage and Current Measurement



connected in parallel

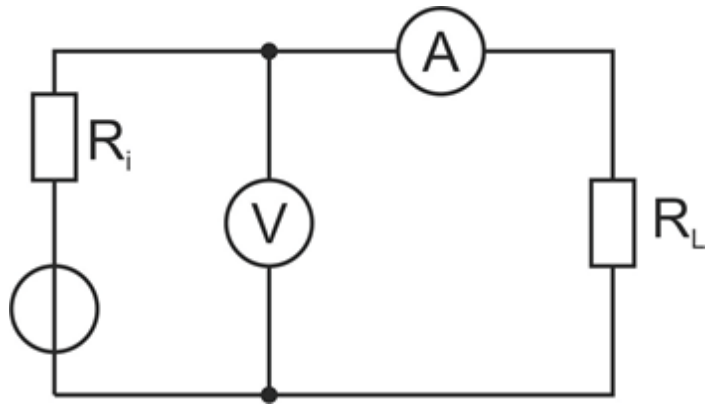
connected in series

ideal:  $R_i = \infty$

ideal:  $R_i = 0$

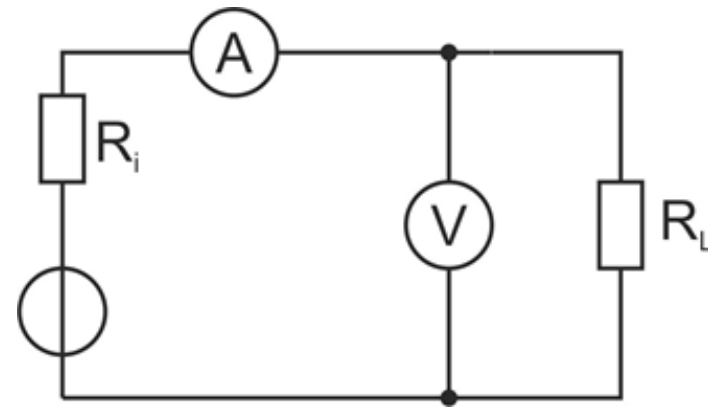
# Simultaneous Measurement

for power measurement: need for simultaneous measurement of voltage and current



measurement correct for current

preferable for high  $R_L$



measurement correct for voltage

preferable for low  $R_L$